Xuan Qi

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Education

University of Washington *Seattle, WA, USA*
Ph.D. Student, Paul G. Allen School of Computer Science & Engineering *Sept 2026 – Present*

- **Research interests:** Self-evolving agent systems, autoresearch agents, agentic reinforcement learning, agent memory, and agent world models.

Tsinghua University *Beijing, China*
Undergraduate, Institute for Interdisciplinary Information Sciences (IIIS, Yao Class; directed by Andrew C. Yao, Turing Award laureate 2000), GPA: 3.85/4.00 *Sept 2022 – June 2026*

- **Coursework:** Natural Language Processing (A^+), Economics and Computation (A^+), Introduction to Programming in C/C++ (A^+), Causal and Statistical Learning (A^+), Mathematics for Computer Science (A), Introduction to Computer Systems (A), Abstract Algebra (A), Probability and Statistics (A), Introduction to Optimization Theory (A), Algorithm Design (A), Machine Learning (A), Game Theory (A)
- **Selected awards:** Yao Award; Tsinghua University Comprehensive Excellence Scholarship; First Prize in the Provincial Division of the Chinese Mathematics Competitions; First Prize in the Provincial Division of the National Olympiad in Informatics in Provinces; First Prize in the Provincial Division of the Chinese High School Mathematics League.

Work Experience

Recursively Self-Improve AI Agent Harness & Posttraining Research Intern *Beijing, China*
Xiaohongshu HI Lab, Pretraining Department (ACE Top Intern Program) *Mar 2026 – Aug 2026*

- Agentic test-time scaling and post-training: Developed efficient test-time scaling methods to generate high-quality agent trajectories. Prepared trajectory-based training data and ran end-to-end supervised fine-tuning (SFT) and reinforcement learning (RL) post-training, with hands-on model deployment, inference optimization, training, and evaluation on models exceeding 200B parameters.
- AI4AI tasks and environments for pretraining improvement: Built tasks and experimental environments in which autoresearch agents propose, implement, and evaluate improvements to the pretraining of their underlying models. Designed controlled experiments and fitted scaling laws to reliably assess these changes, and conducted large-scale pretraining experiments.
- Recursively self-improving autoresearch agents: Built autoresearch agent harness supporting recursive self-improvement across plenty of tasks including MLE-Bench, KernelBench, and nanoGPT; achieved competitive performance across multiple benchmarks.

Algorithm Research Intern *Beijing, China*
Reinforce Omega (AI Agent Start-up) *Sept 2025 – Feb 2026*

- Designed and iterated on *AI agent* workflows, including tool-use policies, tool-creation pipelines, and skill document generation/management to improve robustness and reusability.
- Built context and memory management mechanisms to support long-horizon planning, error recovery, and multi-step execution across heterogeneous tools.
- Contributed to the agent platform backend, including designing agent-friendly websites, improving web/GUI agent capabilities, and creating benchmarks for evaluating agent performance.

Research Intern, Cognitive Computation Group *Philadelphia, PA, USA*
University of Pennsylvania (Advisor: Prof. Dan Roth) *May 2025 – Aug 2025*

- Worked on the *DataProphet* project, analyzing cross-modal transferability of supervision signals across reasoning tasks and, based on these findings, performing real-data reweighting and synthetic-data selection. The resulting data subsets were substantially higher quality than other strong baseline methods, leading models trained under the same budget to clearly outperform baselines and revealing systematic patterns of supervision transfer that improve multimodal LLM data synthesis and training strategies.

Publications

** indicates equal contribution.*

Demystifying Supervision Data Generalization in Multimodal LMs 2026

Xuan Qi, Luxi He, Dan Roth, Xingyu Fu.
ICLR 2026.

Alita: Generalist Agent Enabling Scalable Agentic Reasoning with Minimal Predefinition and Maximal Self-Evolution 2025

Jiahao Qiu*, **Xuan Qi***, Tongcheng Zhang*, Xinzhe Juan, Jiacheng Guo, Yifu Lu, Yimin Wang, Zixin Yao, Qihan Ren, Xun Jiang, Xing Zhou, Dongrui Liu, Ling Yang, Yue Wu, Kaixuan Huang, Shilong Liu, Hongru Wang, Mengdi Wang.
In submission to top conference.

Alita-G: Self-Evolving Generative Agent for Agent Generation 2025

Jiahao Qiu*, **Xuan Qi***, Hongru Wang*, Xinzhe Juan, Yimin Wang, Zelin Zhao, Jiayi Geng, Jiacheng Guo, Peihang Li, Jingzhe Shi, Shilong Liu, Mengdi Wang.
In submission to top conference.

A Survey of Self-Evolving Agents: What, When, How, and Where to Evolve on the Path to Artificial Super Intelligence 2026

Huan-ang Gao*, Jiayi Geng*, Wenyue Hua*, Mengkang Hu*, Xinzhe Juan*, Hongzhang Liu*, Shilong Liu*, Jiahao Qiu*, **Xuan Qi***, …(*Co-first authors ordered alphabetically by last name.*)
TMLR 2026.

One Checkpoint, Many Speeds: Healing Depth Prefixes to Unlock Vertical Compute Control in Reasoning LLMs 2026

Xuan Qi, Yimin Wang.
COLM 2026 Workshop on Efficient Reasoning (**Spotlight**).

PORTAL: Per-User LoRA Memory for Aggregation-Heavy Reasoning at $O(1)$ Context 2026

Xuan Qi, Yimin Wang.
COLM 2026 Workshop on Efficient Reasoning.

Understanding Is Done Early: A Depth Division of Labor in Large Language Models and Its Use for Unbounded-Context Memory 2026

Hanzuo Liu, **Xuan Qi**, Chunyu Liu, Haotian Zhong, Yulong Wang, Rayying, Key, Alex Lamb, Mingyu Gao.
Preprint.

Shallow Preference Signals: Large Language Model Aligns Even Better with Truncated Data? 2025

Xuan Qi*, Jiahao Qiu*, Xinzhe Juan, Yue Wu, Mengdi Wang.
ACL 2025 Workshop.

Difficulty-Based Preference Data Selection by DPO Implicit Reward Gap 2025

Xuan Qi, Rongwu Xu, Zhijing Jin.
In submission to top conference.

Proper Dataset Valuation by Pointwise Mutual Information 2026

Rui Ray Chen*, **Xuan Qi***, Yuxin Chen, Yongchan Kwon, James Zou, Shuran Zheng.
EC 2026; earlier version at ICLR 2025 Workshop.

DebateQA: Evaluating Question Answering on Debatable Knowledge 2026

Rongwu Xu*, **Xuan Qi***, Zehan Qi, Wei Xu, Zhijiang Guo.
Findings of EACL 2026.

EdgeBench: Unveiling Scaling Laws of Learning from Real-World Environments 2026

Deyao Zhu, Xin Zhou, Shengling Qin, Xuekai Zhu, Hangliang Ding, Shu Zhong, Zixin Wen, Zhonglin Xie, Chenhui Gou, Linxuan Ren, Yueyang Wang, Junfeng Zhong, Rui Liu, Tian Gao, Yangguang Lin, Jingyuan Zhang, Maojia Song, **Xuan Qi**, et al.
Preprint.

Learning Agent Routing From Early Experience 2026

Yimin Wang*, Jiahao Qiu*, **Xuan Qi**, Xinzhe Juan, Jingzhe Shi, Zelin Zhao, Hongru Wang, Shilong Liu, Mengdi Wang.
In submission to top conference.

On Path to Multimodal Historical Reasoning: HistBench and HistAgent 2025

Jiahao Qiu*, Fulian Xiao*, Yimin Wang*, Yuchen Mao*, Yijia Chen*, Xinzhe Juan, Siran Wang, **Xuan Qi**, Tongcheng Zhang, Zixin Yao, et al.

In submission to top conference.

AgentDistill: Training-Free Agent Distillation with Generalizable MCP Boxes 2025

Jiahao Qiu*, Xinzhe Juan*, Yimin Wang*, Ling Yang*, **Xuan Qi**, Tongcheng Zhang, Jiacheng Guo, Yifu Lu, Zixin Yao, Hongru Wang, Shilong Liu, Xun Jiang, Liu Leqi, Mengdi Wang.

In submission to top conference.

HDI4D: A Hand-Deformable Object Interaction Dataset for Soft Body Hanging 2024

Pengyang Shi*, Xiaoxi Liang*, Xiaomeng Xu, **Xuan Qi**, Tongzhou Du, Yi Li.

Research Experience

Wang Lab@Princeton AI for Accelerating Invention *Remote*

Princeton University (Advisor: Prof. Mengdi Wang) *Nov 2024 – Feb 2026*

- Co-led the development of *Alita*, a general agent with minimal pre-definition and maximal evolution that evolves via dynamic Model Context Protocol (MCP) creation and achieved top-1 performance on the GAIA benchmark.
- Extended the Alita framework to *Alita-G*, which systematically abstracts and curates MCP tools to transform generalist agents into domain experts and provide a principled pathway from generalist capabilities to reusable, domain-specific competence.
- Served as co-first author of *A Survey of Self-Evolving Agents*, mapping the emerging landscape of self-evolving architectures and agentic systems.
- Served as co-first author of *Shallow Preference Signals: Large Language Model Aligns Even Better with Truncated Data?*, showing that truncated preference signals can effectively steer model behavior and revealing that models often rely on shallow heuristics for alignment.
- Contributed to *AgentDistill*, exploring training-free distillation via Model Context Protocols to transfer reasoning and tool-use abilities between agents.

EconCS Lab *Beijing, China*

Tsinghua University, IIIS (Advisor: Assistant Prof. Shuran Zheng) *Jul 2024 – Jan 2025*

- Developed a metric based on pointwise mutual information to value datasets in machine learning and encourage honest data contributions, and carried out theoretical analyses and experiments to validate its effectiveness in *Proper Dataset Valuation by Pointwise Mutual Information*.

The 2nd Robot Dog Competition *Beijing, China*

Tsinghua University, Department of Automation *Apr 2024 – Jul 2024*

- Trained Xiaomi's CyberDog2 to play soccer: trained the robot dog to perform locomotion and dribbling in a simulated environment using reinforcement learning, combined the two using a rule-based approach, and utilized interference frequency optimization, default pose adjustment, and PD control parameter tuning to solve the sim-to-real gap.
- Our project got the Best Technology Award of the 2nd Robot Dog Competition organized by the Department of Automation.

3DVICI Lab *Beijing, China*

Tsinghua University, IIIS (Advisor: Assistant Prof. Yi Li) *Aug 2023 – Oct 2024*

- Spearheaded the data pipeline for *HDI4D: A Hand-Deformable Object Interaction Dataset for Soft Body Hanging*, designing robust data-collection pipelines, managing complex annotations, and empirically validating models to support robotic understanding of deformable object interactions.

Awards and Honors

- Yao Award (10 out of 96), 2025.
- Tsinghua Friends – Huawei Scholars First-Class Scholarship (10 out of 96), 2025.
- Science and Technology Innovation Excellence Scholarship, Tsinghua University (10 out of 96), 2025.
- Outstanding Volunteer and Public Service Scholarship, Tsinghua University (5 out of 96), 2025.
- Tsinghua University Comprehensive Excellence Scholarship (15 out of 96, awarded to those with outstanding performance), 2023, 2024.
- Tsinghua Friends – Zhang Mingwei Scholarship (10 out of 96), 2023.
- Tsinghua Friendship – Huang Yicong and Spouse Scholarship (10 out of 96), 2024.
- Best Technology Award of the 2nd Robot Dog Competition organized by the Department of Automation (1 out of 50 teams), 2024.
- First Prize in the Provincial Division of the Chinese Mathematics Competitions, 2023.
- Outstanding Student of Shandong Province (High School), 2021.
- First Prize in the Provincial Division of the Chinese High School Mathematics League, 2021.
- First Prize in the Provincial Division of the National Olympiad in Informatics in Provinces (NOIP), 2020.
- Second Prize in the National High School Physics Competition, 2021.
- Second Prize in the Provincial Division of the National High School Chemistry League, 2021.
- Second Prize in the Provincial Division of the National Olympiad in Informatics in Provinces (NOIP), 2019.

Leadership and Service

- Reviewer for ICLR 2026 and 2027, NeurIPS 2026, ICML 2026, COLM 2026, and AAAI 2026; ACL Rolling Review (ARR), 2025–2026, including the May and August 2026 cycles.
- Served as the team leader of the Tsinghua University "Love for Alma Mater" event's Jining Confucius School team, and as a key volunteer for the Tsinghua University Jining Regional Admissions Group in 2023 and 2024.
- Participated in multiple rural teaching-support programs organized by the Tsinghua University SAEPA (Student Association for Education and Public Affairs), including classroom teaching and one-on-one mentoring for high school students in under-resourced areas.
- Tsinghua University Q&A Forum Senior Volunteer.
- Tsinghua University Three-Star Zijing Volunteer.
- Served as the Class Publicity Officer, responsible for designing and promoting class activities.

Skills

- **Large-scale model deployment and inference:** Deploying and accelerating inference for open-weight models, including Kimi K3, GLM 5.2/5.3, and DeepSeek V4/V4.1.
- **Large-scale model training:** Hands-on training experience with Qwen3.5 (35B), Qwen3 (235B), and Xiaohongshu's large-scale internal models, including **200B+ models**; pretraining experiments, scaling-law fitting, and end-to-end SFT/RL post-training from data preparation through training and evaluation.
- **Programming Languages:** Python, C/C++, JavaScript.
- **Tools and Platforms:** Git, LaTeX, SQL, Xiaomi CyberDog2, Raspberry Pi.
- **Languages:** Chinese, English.
- Other: Passionate Go player; achieved third place in the *San Yi Cup* National Dan Promotion Tournament and ranked in the top ten multiple times in the Shandong Youth and Children Championships.